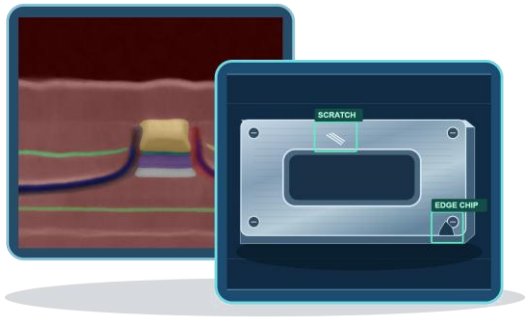
HawkeyePlus Platform Architecture

Connect industrial images, model training, validation and release in one platform

01 Data & Annotations



Images + Labels / Masks

Import · Annotate · Preview

Visual Training Hub

Dataset · Model · Training



Job Setup · GPU Selection · Training Monitor

02 Validation & Delivery

Evaluate & Infer
Metrics · Prediction preview

Model Export
Convert · Package · Download

Model Registry
Product versions · Activation

Multi-framework Training

PaddlePaddle
Detect · Classify · Segment

PyTorch
Detect · Segment

anomalib
Anomaly Detection

Compute & Asset Foundation

Compute
GPU / Python

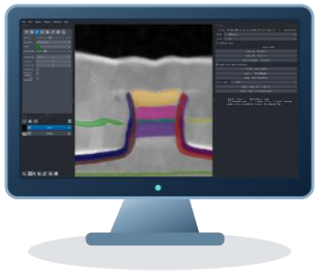
Data & Assets
Images · Weights · Logs

- **Management** Project ownership · User roles · Environment checks · GPU status · Storage quotas · Activity logs

A Visual Lifecycle: From Data to Models

Bring preparation, training and validation into a shared model-development workflow

01 Prepare Data



Import images & labels

Convert formats · Review samples

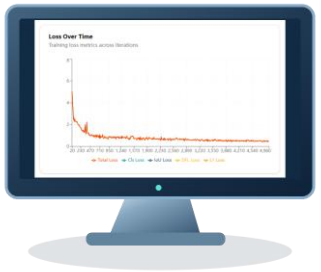
02 Configure



Dataset + Model + Parameters

Reusable templates · Visual setup

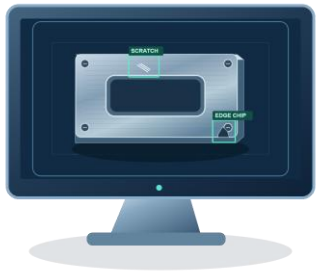
03 Training Monitor



Run jobs · Track training

Progress · Loss · Metrics

04 Validate



Evaluate · Preview predictions

Inspect results · Assess readiness

05 Deliver Models



Export · Register versions

Select the active product model

Use validation results to refine data and configuration

Four vision tasks. One shared training platform.

Object Detection
Locate objects and defects

Image Classification
Recognize categories and quality

Semantic Segmentation
Map regions and boundaries

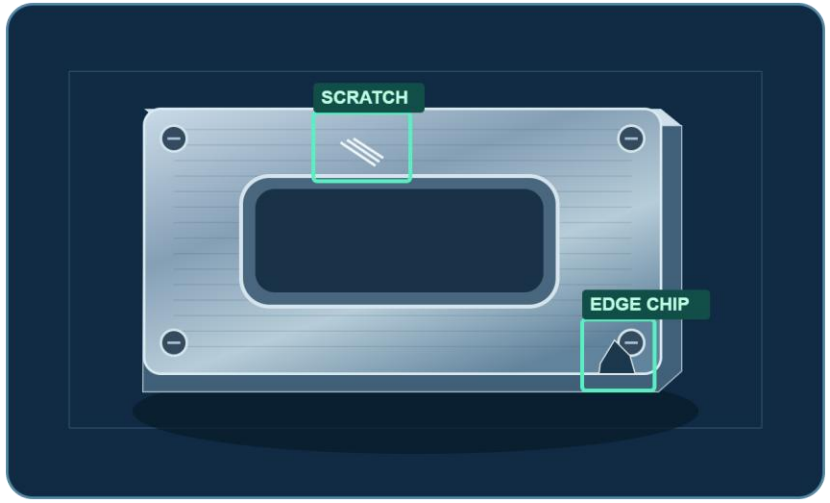
Anomaly Detection
Learn from normal samples

Simplify adoption / Reuse training knowledge / Build model assets

Object Detection | Industrial Defect Localization

Locate defects and identify their classes with bounding boxes and confidence scores

Locate & Classify Defects



Output: Boxes · Classes · Confidence

Illustrative output, not a model prediction

PaddleDetection

7 presets

PP-YOLOE · YOLOv3 · PicoDet · RT-DETR

Faster R-CNN · RetinaNet · CenterNet

TorchDet

4 presets

Faster R-CNN · RetinaNet · FCOS · SSD / SSDLite

SSD variants: VGG16 → SSD300; MobileNetV3 → SSDLite320



COCO Images & Labels

Box labels · Classes



Training & Metric Tracking

mAP / mAP50 · Training curves



Detection Preview

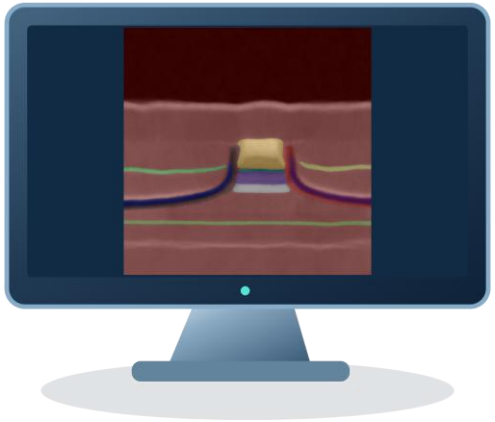
Locate defects · Inspect classes

Presets are backend-specific; PP-YOLOE and RT-DETR are not TorchDet models.

Segmentation Models | Pixel-level Understanding

TEM images illustrate segmentation across compact, encoder-decoder and Transformer architectures

Pixel-level Prediction Example



Regions · Boundaries · Label masks

TEM segmentation example from this project

PaddleSeg

7 presets

PP-LiteSeg · BiSeNetV2 · OCRNet · UNet · DeepLabV3+
SegFormer-B0 · FCN

TorchSeg

4 presets

UNet · DeepLabV3 (UI: DeepLabV3+) · FCN · LR-ASPP

TorchSeg maps DeepLabV3P to torchvision DeepLabV3, not a V3+ decoder.



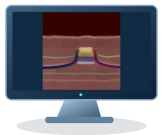
Dataset Layout

Images + masks + train / validation splits



Evaluation Metrics

mIoU · Dice · Accuracy



Applications

Defect regions · Microstructure · Boundaries

Classification & Anomaly Detection

From image-level labels to learning normal appearance, reuse the same model-development workflow

PaddleClas | Image Classification

Learn labeled categories; predict image-level classes



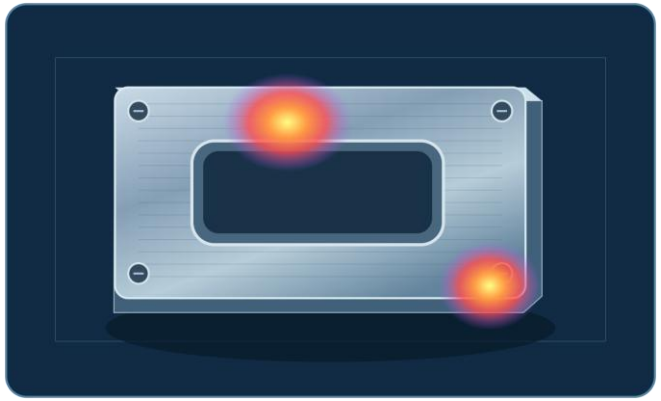
Illustrative classification

9 Classification Variants

ResNet18 · ResNet34 · ResNet50 · ResNet101
MobileNetV3 Small / Large (x1.0)
PP-LCNet x1.0 · PP-HGNet Small
Swin Transformer Tiny (patch4 / window7 / 224)

TorchAnomaly | Anomaly Detection

Learn normal appearance; produce anomaly scores and heatmaps



Illustrative anomaly heatmap

5 anomalib Presets

PatchCore · PaDiM · STFPM · EfficientAD

SuperSimpleNet

Current UI presets only; custom YAML compatibility depends on the selected backend.